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AI X Seeds of Promise

What you'll learn today

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1. The story of AI so far
2. How AI models actually work
3. How to use AI
4. Case studies


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The story of
AI so far

The story so far

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2017: FIRST MAJOR BREAKTHROUGH

2017

The Transformer

Google creates the **transformer**
a new neural network architecture

Why is this breakthrough significant?

1. Transformers are much better at **understanding the relationships** between inputs into a model (e.g the order and weight of each word in a sentence)

2. More generalisable and able to handle lots of different tasks, vs previous AI systems that were built with a single task in mind

Key takeaway

2017 flipped the global switch: AI growth surged, flooding every field with **possibility**

The story so far

2017-2022: EXPONENTIAL AI DEVELOPMENT

2018

GPT-1

Where is the Eiffel Tower? X

'The book is on the table' → 'El libro es en la tabla' X

Trained on **7,000** books / **5GB** data

117m parameters

2019

GPT-2

Trained on **240GB** of data

1.5bn parameters

2020

GPT-3

Trained on **45TB** of data - almost the size of the entire internet

175bn parameters

Key takeaway

After half a decade of rigorous advancement, **2022** witnessed **ChatGPT's launch**

The story so far

2022-2024: WHERE IS GPT-5?

2023

GPT-4

~10x increase in parameters; meaningfully better performance

Scaling laws seem to be **holding**

GPT-4 introduces **multimodal;** to create **images,** & use images as part of the prompt

2024

Pivotal moment

By **mid-2024,** something **shifted;** GPT-4 came out March 2023, GPT-5 is not out

The feeling is that **scaling laws** are starting to **break**... there is **little data left** available to train on

PLUS

Competition is heating up:

Anthropic:
Claude 3.5 & 3.7

Google:
Gemini 2.5

Meta:
Llama 3

The story so far

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2024: THE FIRST REASONING MODEL

2024

GPT-4

o1 is out

Rather than ramping up training, **o1** is built around training using **chain of thought reasoning at inference time** (i.e when generating it's answer)

- It generates tokens e.g. 'Let's think step by step…', then uses output of that to generate final answer

Key takeaway

Reasoning models have arrived, ushering in a game-changing leap in AI power & versatility

Meaningfully **better results**, and less hallucinations

- In the International Math Olympiad qualifying exam, **o1** answered **83%** of the questions correctly, while **GPT-4o** answered **13%** correctly

The story so far

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DON'T THINK OF IT AS PRE & POST-CHAT GPT

Performance has improved massively since ChatGPT was released:

What's possible today is order of magnitudes **more complex** than what was possible in November 2022 when ChatGPT was released

Iterative progression in the release of GPT models:

GPT-3 / GPT-4
o1 / o3
Claude, Gemini etc

Combined with tooling and functionality:

Image generation
Deep research
Web search
MCP (integrations into other apps)

Key takeaway

Overall trend: every release is **smarter**, **faster**, and **cheaper** to run


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How AI models
actually work

How LLM models work

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Simplistically: Trained on huge datasets to do next word prediction

In more detail...

It doesn't just look one word ahead

An AI can **peek a few steps ahead** - like deciding the final rhyme of a poem before filling in the line before - so it **doesn't write strictly one word at a time**

It solves problems in unique ways

When you ask it to add 37 + 48, it **isn't doing schoolbook arithmetic.** Instead, it **relies on patterns** it saw in billions of examples of numbers and answers that happen to add up correctly

Superpower = pattern-matching creativity

Give it tons of recipes, songs, or legal briefs and it can **blend those patterns into something new**, often surprisingly clever

How LLM models work

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Simplistically: Double checking your answers is really important

Weak spots...

Lack of Interpretability:
We can't see how it gets from A to B

We can't see inside. It's hard to explain exactly why it chose a particular answer - **its reasoning is hidden in sprawling math**

Lack of Determinism:
Its processes are probability based rather than rule based

Its not perfectly rigid. For crystal-clear, one-right-answer tasks (e.g. 2+2=4) it can still slip up - **its process is probabilistic rather than strictly rule-based**

Example model

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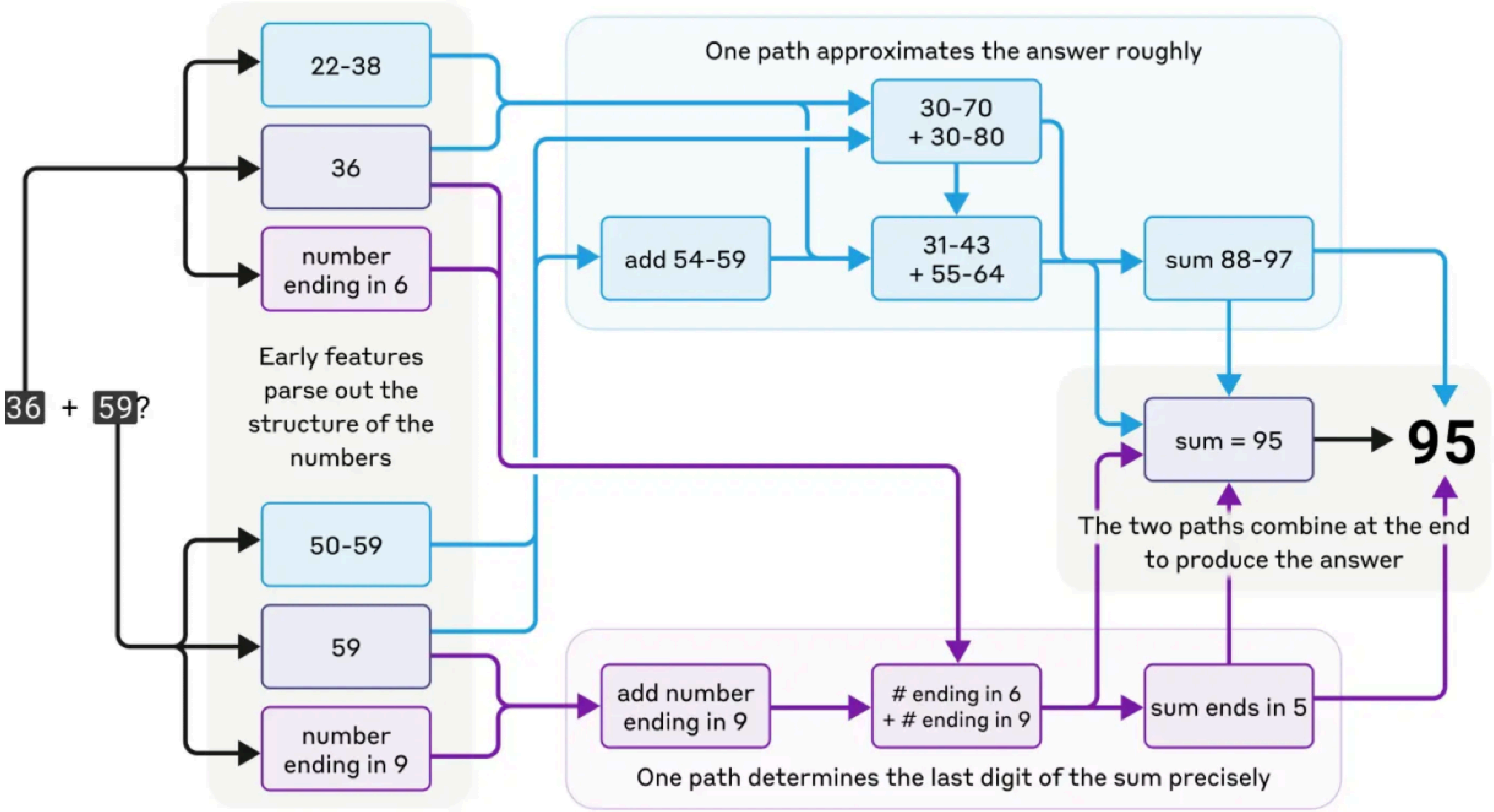


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How to
use AI
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Context in Prompt Engineering

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C-R-A-F-T

C-ontext
(facts you know)

R-ole
(who the AI should be)

A-ction
(the task)

F-ormat
(table, bullet list, script, etc.)

T-one
(clear, simple, stern, friendly, excitable)

Context & Prompt engineering

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Give the AI the right context, be specific, and use prompt engineering techniques

- The difference between a good answer and a bad answer is often down to the **context you give it**

- When you are prompt engineering, you are trying to shift the probabilities of the next word towards the one you want
  - Use words like ‘**never**’ rather than ‘not’,
  - ‘**exactly**’ rather than ‘like’,
  - say **‘It is really important that…’**
  - Get it to take the task more seriously by saying **‘This is a really important task that I want you to take really seriously’**
  - **‘Give me specific sources for everything you provide’**
  - **‘What context should I give you to allow you give me a better answer here?’**
- Using cheaper, quicker models to **build a prompt** for slower, reasoning models is a great trick